

collector and emitter terminals, it is necessary to bias the device as shown in Fig. 7(a). The injector power supply V_{JJ} determines the voltage range of the negative resistance. The collector may then be biased so that the quiescent voltage $V_{CE} = \frac{1}{2}V_{JJ}$. This places the operating point in the middle of the negative-resistance region, thus allowing a symmetrical operation. For stability of the operating point it is required that the collector-biasing resistance be less than the magnitude of the negative resistance. The base current determines the valley point of the negative-resistance characteristic.

The value of the negative resistance can be varied over a wide range by simply varying R_i . It should be noted that as R_i decreases, the peak point is raised since I_J [see (8)] increases, and vice versa. The minimum value of negative resistance is found from (9) with $R_i = 0$ as $|(R_n)_{\min}| = h_{32}/h_{f1}$, but this is the least stable, since it depends solely on device parameters.

VI. CONCLUSIONS

Several linear applications of a p-n-p-n tetrode have been described. A simple amplifier stage using a 3N81 device operating under small-signal conditions can provide over 40 dB voltage-gain variation for an injector current variation typically from 10 to 180 μ A. The dependence of

the voltage gain on the injector current is linear in a wide region and this makes possible the operation of the tetrode as a signal multiplier and as an amplitude modulator. The maximum modulation percentage attainable with negligible envelope distortion is in the order of 50 percent. The negative resistance obtainable is stable, essentially pure up to about 50 kHz (for the device investigated), and conveniently controllable over the range from about 100 ohms to megohms. The current and voltage ranges of the negative-resistance region are widely adjustable by external dc sources.

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Feedback Stabilized Four-Quadrant Analog Multiplier

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Abstract—A technique utilizing an ideal controlled current fork as a fundamental building block is evolved to generate the product of two variables. A particularly useful implementation of the technique, making use of the exact exponential characteristics of the emitter-base junction of the bipolar transistor to realize a controlled current fork, is demonstrated. It is shown that a mismatch of transistor pairs and a mismatch of resistance values give rise to certain predictable errors for which quantitative expressions are derived in order to determine acceptable tolerances.

Measurements on a particular design are given to support the theoretical derivations, and the operation of the multiplier as a squarer, independent of transistor-pair mismatch, is demonstrated.

I. INTRODUCTION

WHENEVER the need for a reasonably accurate electronic analog multiplier has arisen in the past, systems designers have been generally compelled to choose from the following two types.

1) The quarter-square multiplier [1], [2] that utilizes two square-law devices to obtain the product of two input quantities X and Y according to the equation that follows.

$$4XY = (X + Y)^2 - (X - Y)^2.$$

2) The time-base multiplier [1], [2] that operates on the principle that when a pulse is amplitude modulated proportional to one input quantity and width modulated proportional to the other input quantity, the area of the pulse is a direct measure of the product of the two input quantities and may be obtained by measuring the mean value of a train of such pulses.

Commerically available analog multipliers have been generally closely related to one or the other of these two types. Their static errors are typically of the order of a fraction of a percent, the time-base type being usually superior in this respect. Bandwidths are of the order of tens of kilohertz—the quarter-square type generally giving a better performance.

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Implementation of either of the above techniques to realize an analog multiplier having a given performance can be an extremely tedious task, frequently resulting in fairly complex hardware. The difficulty encountered in their design is due to the necessity of using some non-linear element or process (as implied by the principle of superposition). The accuracy of a multiplier is limited by the deviation of the behavior of the nonlinear element from its ideal behavior.

Mention must also be made of another type of analog multiplier that finds its application where high-static accuracies are required and speed of operation is of secondary interest. This is the servomultiplier [9] that uses a slave multiplier (usually a motor-driven potentiometer) to linearize the output with respect to one or other of the input variables.

In the past few years, a number of techniques utilizing the very predictable exponential characteristics of the emitter-base junction of the bipolar transistor [3], [8] have been exploited to generate the product of two variables. One such technique [4], is based on the slide-rule principle, whereby logarithms and antilogarithms are obtained directly. Frater [5] and Morton [6] make use of the small-signal characteristics of the bipolar transistor.

These transistor multipliers all suffer from at least one of the following limitations:

- 1) dependence on transistor parameters;
- 2) nonlinear behavior with respect to one or both inputs;
- 3) temperature dependence of the product coefficient;
- 4) low signal-to-noise ratios at low output levels;
- 5) limited bandwidth;
- 6) excessive dc drift on the output.

Recently, a transistor multiplier operating according to the "current-rationing" principle has been proposed [7]. The principle may be considered as a modification of the technique described in [5], [6] and results in a linearizing of the transfer response at the extremities of the dynamic range as well as overcoming the other problems listed above. Bandwidths of above 10 MHz have been achieved using this technique.

In the following sections, a feedback technique that realizes an analog multiplier is presented and the suitability of the bipolar transistor for its implementation is demonstrated. Furthermore, it is shown that the disadvantages listed above are overcome. Also, the constant of multiplication is shown to depend primarily on the values of passive circuit elements (resistances) and seems to be analogous to expressions obtained for gain of modern feedback stabilized amplifiers in the realm of linear circuits.

II. GENERAL THEORY

The description of the multiplier presented here is generalized by introducing a fictitious "ideal controlled current fork" having, as yet, no particular relation to any existing devices. A schematic of this fundamental

building block is shown in Fig. 1(a). An input current I_{in} is divided into two components αI_{in} and $(1 - \alpha)I_{in}$, where α is determined solely by the controlling parameter χ .

Let us now consider an n -stage ladder consisting of controlled current forks as shown in Fig. 1(b). The feed to the first stage is a constant current I_0 .

The input of the j th stage is $I_{j-1} - I'_{j-1}$ and its output $I_j - I'_j$. The two current forks of the j th stage are both controlled by the parameter χ_j (which may be a voltage or a current) and have, therefore, both a current-splitting ratio of α_j ; k_0 is a current attenuation constant.

The current forks are connected so that the following relationships exist:

$$\begin{aligned} i_A &= \alpha_j I_{j-1} \\ i_B &= (1 - \alpha_j) I_{j-1} \\ i_C &= (1 - \alpha_j) I'_{j-1} \\ i_D &= \alpha_j I'_{j-1}. \end{aligned}$$

Therefore, for this stage:

$$\begin{aligned} I_j - I'_j &= k_0(i_A - i_B + i_C - i_D) \\ &= k_0(I_{j-1} - I'_{j-1})(2\alpha_j - 1) \end{aligned} \quad (1)$$

and, by induction, for the n stages:

$$I_n - I'_n = (k_0)^n I_0 \prod_{j=1}^n (2\alpha_j - 1). \quad (2)$$

From (1) it is apparent that the output of the j th stage is a product of the variables $I_{j-1} - I'_{j-1}$ and $2\alpha_j - 1$, which is a function (in general, nonlinear) of χ_j . The range of the term $2\alpha_j - 1$ will determine the operating mode of the multiplier; for four-quadrant operation $2\alpha_j - 1$ must be variable about zero.

Assuming that the quantity $I_{j-1} - I'_{j-1}$ can be linearly controlled, the problem that remains (in obtaining linear control of the output with respect to both inputs) is one of linearizing the term $(2\alpha_j - 1)$ with respect to some input parameter, i.e., i_j .

The problem can be solved if we can find a *linear* combination of the currents i_A , i_B , i_C , and i_D in such a way that they are simply related to the quantity $(2\alpha_j - 1)$ alone. That is, if the parameter $(2\alpha_j - 1)$ can be sensed somewhere after the control process (current splitting), then this sensed quantity may be compared with the current i_j in a high-gain difference amplifier (whose output is the controlling parameter χ_j) in order to affect the tracking of this sensed quantity and i_j . In order to ensure the validity of such a feedback technique, the dependence of α_j on χ_j shall be restricted to monotonically increasing functions. The required linear combination of the currents i_A , i_B , i_C , and i_D is in fact:

$$i_A - i_B - i_C + i_D = (I_{j-1} + I'_{j-1})(2\alpha_j - 1) = I(2\alpha_j - 1) \quad (3)$$

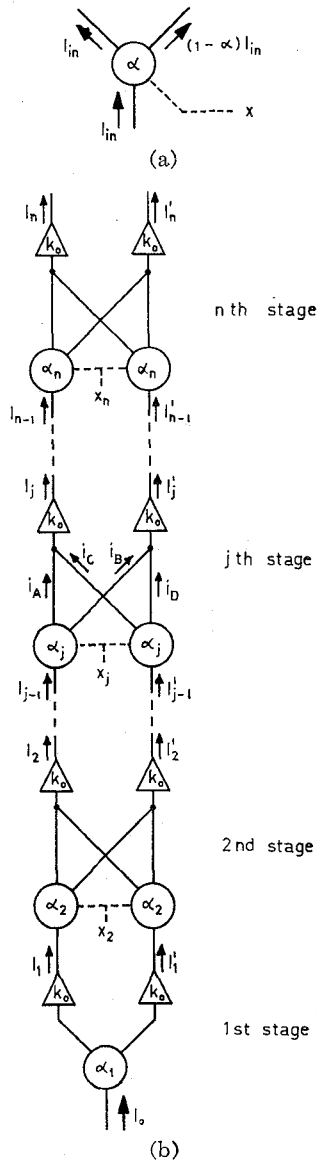


Fig. 1. (a) Ideal controlled current fork; (b) multiplying ladder utilizing controlled current forks.

where $I = (k_0)^j I_0$ is constant. If then, a proportion of this quantity $i_F = k_F I (2\alpha_i - 1)$, where k_F is constant is compared with i_i in a current difference amplifier, of transresistance Z whose output is a controlling voltage, i.e., x_i , we have for the closed loop of Fig. 2:

$$I(2\alpha_i - 1) = \frac{1}{k_F} \left(\frac{x_i}{Z} + i_i \right). \quad (4)$$

For sufficiently large values of Z , (4) simplifies to

$$2\alpha_i - 1 \approx \frac{i_i}{k_F I}. \quad (5)$$

Substituting (5) in (1) we obtain an expression for the linearized multiplier stage:

$$I_i - I_i' = \frac{k_0}{k_F I} i_i (I_{i-1} - I_{i-1}'). \quad (6)$$

In deriving (6), the following assumptions have been

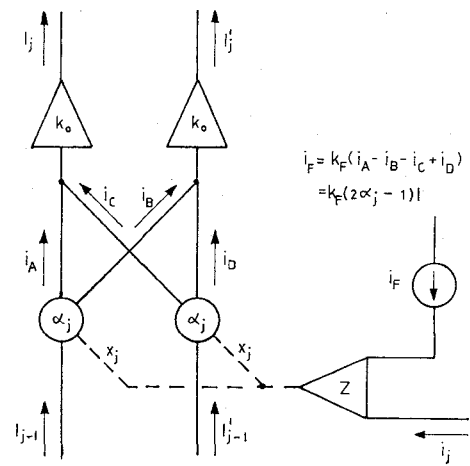


Fig. 2. Basic multiplier configuration with the feedback technique.

made: 1) the current forks are ideal; 2) the two current forks of each stage are identical (i.e., their values of current-splitting ratio α_i are always equal).

It shall be shown in the next section, in which a particular implementation of the technique using bipolar transistors is considered, that the second of these assumptions may not always be valid, and that a deviation from the ideal behavior gives rise to predictable errors.

III. IMPLEMENTATION USING THE BIPOLAR TRANSISTOR

For the implementation of the described technique we must find a realization of our fundamental building block—the controlled current fork. One transistor configuration, which makes use of the exact exponential characteristics of the bipolar transistor to achieve the required operation, is shown in Fig. 3.

It shall be assumed that the transistors are operated in a range in which the collector current is (to a very good approximation) dependent exponentially upon the base-emitter voltage, namely,

$$i_C = \frac{1}{1 + \frac{1}{\beta}} I_{C0} \exp \left(\frac{q v_{BE}}{K T} \right) \quad (7)$$

where

- i_C collector current,
- v_{BE} base-emitter voltage,
- I_{C0} base-emitter reverse bias saturation current,
- q electronic charge,
- K Boltzmann's constant,
- T absolute temperature,
- β base-collector current amplification factor.

For the sake of simplicity, the factor $1/(1 + (1/\beta))$ shall be taken as unity for all transistors in the following analysis. (In the final expression for the output of the stage, the factor $1/(1 + (1/\beta))$ is included to represent this current loss.)

For the configuration shown in Fig. 3 it can be shown that

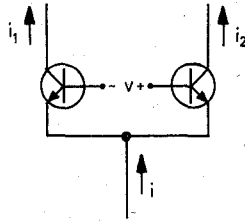


Fig. 3. Implementation of a controlled current fork utilizing bipolar transistors.

$$\frac{i_1}{i} = \alpha = \frac{1}{1 + \exp \frac{q(v + \Delta v)}{KT}} \quad (8)$$

and therefore,

$$2\alpha - 1 = \tanh \left[\frac{-q(v + \Delta v)}{2KT} \right] \quad (9)$$

(v here is analogous to χ in Fig. 1(a)) where Δv is the base-emitter mismatch voltage of the transistor pair for equal emitter currents (i.e., for $\alpha = 0.5$).

An examination of (8) shows it to be of the required form (i.e., monotonically increasing) for the correct operation of the feedback technique.

A number of circuit configurations using such a current fork exist. Perhaps the simplest of these is that shown in Fig. 4. In this circuit, each current-splitting operation is performed with a transistor pair; however, due to the unequal mismatches of the pairs, the two current forks may no longer be considered to have identical values of current-splitting ratios over their operating ranges.

The two distinct values of current-splitting ratio are now α_i and α'_i such that:

$$2\alpha_i - 1 = \tanh \left[\frac{-q(v + \Delta v)}{2KT} \right] \quad (10)$$

$$2\alpha'_i - 1 = \tanh \left[\frac{-q(v + \Delta v')}{2KT} \right] \quad (11)$$

where

Δv = base-emitter mismatch voltage of left-hand pair,
 $\Delta v'$ = base-emitter mismatch voltage of right-hand pair.

By making suitable approximations we may derive the following useful expression (see Appendix I) relating the normalized output quantity $Z = (I_i - I'_i)/(k_o I)$ to the normalized input quantities $X = (I_{i-1} - I'_{i-1})/I$, $Y = i_i/(k_F I)$, and the mismatch parameter $(\Delta v - \Delta v')$:

$$Z = \gamma' \left[XY + \frac{q(\Delta v - \Delta v')}{4KT} (X^2 + Y^2 - 1) \right]. \quad (12)$$

Note that

$$k_o = \frac{R_F}{R_C + R_F}$$

$$k_F = \frac{R_C}{2(R_C + R_F)}$$

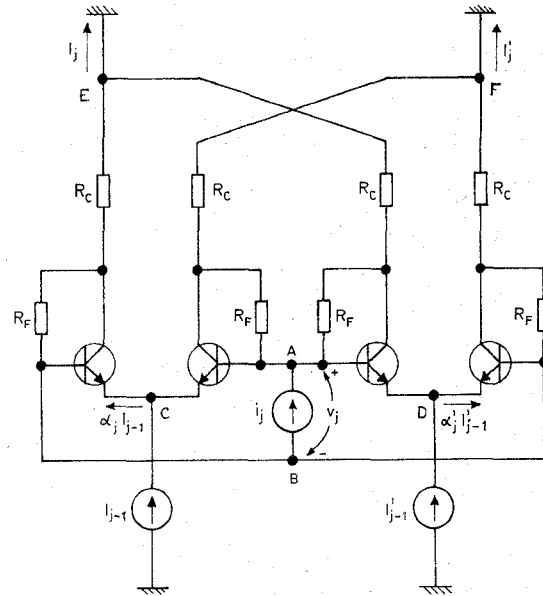


Fig. 4. Implementation of a multiplier stage utilizing bipolar transistors.

(see Appendix I) where

$$\gamma' = \left[\frac{1}{1 + \frac{1}{\beta}} \right] \left[\frac{1}{1 + \frac{1}{2k_F \beta}} \right].$$

In most practical cases, the value of γ' will be very nearly equal to unity, giving us the very useful design formula for $\Delta v = \Delta v'$:

$$I_i - I'_i \approx -\frac{2R_F}{R_C I} [i_i(I_{i-1} - I'_{i-1})]. \quad (13)$$

It may be shown that the small-signal driving-point impedance across the nodes C and D (Fig. 4) is approximately equal to $(4KT)/qI$. Also the driving-point impedance across the nodes A and B may be shown to be approximately equal to $(4KT)/qI[(R_C + R_F)/R_C]$. In a satisfactory design these input resistances are much smaller than either R_C or R_F —and an extremely simple way of cascading such stages to realize multiple product multipliers or polynomial generators is apparent. By letting the nodes E and F of a particular stage be identical to the nodes C and D (or A and B), respectively, of the following stage, the desired current feed into that stage is achieved due to the gross impedance mismatch at these nodes and the desired functional ladder may be synthesized by stacking the stages in such a manner.

In the derivation of the expression for the output of the stage, $I_i - I'_i$ as given by (12), a number of approximations were made; however, the accuracy of the expression (from an engineering design point of view) is borne out by measurement.

It is apparent that the suitability of the circuit as an analog multiplier is dependent primarily on the degree of mismatch of the transistor pairs, the unwanted terms being linearly dependent upon the quantity $(\Delta v - \Delta v')$.

A quantitative comparison of the error terms arising

in the above multiplier (having a given mismatch voltage) with that arising in the multiplier proposed by Gilbert [7] (in which two distinct mismatch voltages must be considered) is given in Appendix II.

The accuracy of (12) is greatest for $\alpha_i \approx 0.5$ (i.e., when all current-fork transistors carry approximately equal currents). As the value of α_i approaches zero or unity, the loop gain that stabilizes the value of $(2\alpha_i - 1)$ with respect to i_i decreases. The reason for this decrease is that i_F is supplied through a resistive mesh (R_C and R_F) and is not an ideal current source. Measurements on satisfactory designs, have indicated no appreciable distortion occurs at the extremities of the dynamic range due to this effect.

IV. PRACTICAL MULTIPLIER

The single-stage multiplier (shown in Fig. 5) was developed for use in an accurate audiocorrelator and served to verify the above theory.

Inputs $I_{i-1} - I'_{i-1}$ and i_i are derived from balanced current sources driven by input voltages v_1 and v_2 , respectively. The output of the stage is taken in the form of a balanced voltage v_3 .

Symmetrical operation with respect to both inputs is achieved by designing the current drives such that:

$$\frac{I_{i-1} - I'_{i-1}}{2i_i} = \left(\frac{R_C + R_F}{R_C} \right) \frac{v_1}{v_2}. \quad (14)$$

Biasing of the current-dividing transistors Q_1 , Q_2 , Q_3 , and Q_4 is achieved by maintaining a quiescent current through the resistances R_F . The quiescent currents in transistors Q_1 , Q_2 , Q_3 , and Q_4 were set as low as $100 \mu A$ (in order to reduce the effects of high-level injection and the ohmic component of base resistance, and to keep the power dissipation low enough to ensure a constant value of q/kT). The quiescent emitter-collector potentials are 6 volts.

In order to verify the foregoing theory, the operation of the circuit was determined using transistor pairs having varying degrees of mismatch as shown in the following tabulation.

Pair Number	Δv (mV)	$\Delta v'$ (mV)	$ \Delta v - \Delta v' $ (mV)
1	+ 2.25	+ 2.24	0.01
2	+ 0.26	+ 0.16	0.10
3	+ 0.43	+ 0.17	0.26
4	+ 1.10	+ 2.30	1.20
5	+ 2.50	+ 1.10	1.40
6	+ 1.13	- 1.70	2.83
7	+ 2.25	- 2.24	4.49

A. Linearity

A plot of static linearity using transistor pair 2 is shown in Fig. 6. The measured value of the product coefficient was within 7 percent of the predicted value given by (13) and within 2 percent when the correction factor γ' was taken into account.

Agreement with the behavior predicted by (12) is illustrated in Fig. 7(a) and (b), which show logarithmic

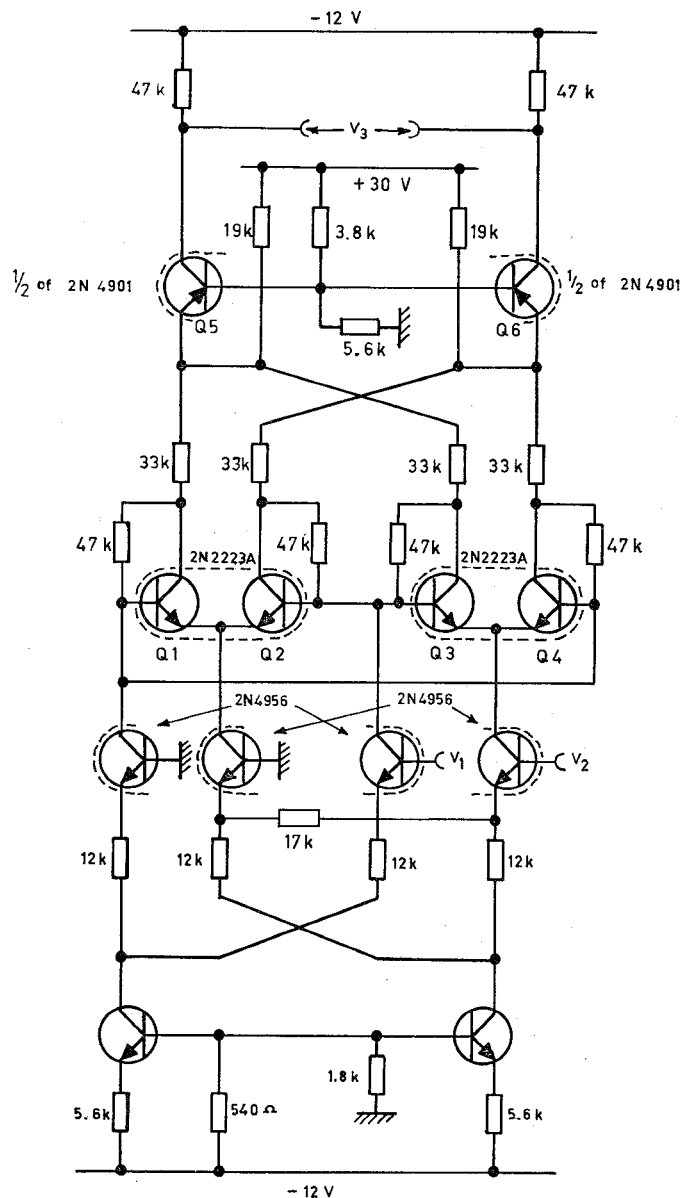


Fig. 5. Complete realization; v_1 and v_2 are input voltages; v_3 is the output voltage.

plots of one input as a function of the second input with the output held constant at 100, 10, 1, 0.1 percent, and zero, based upon 100 percent = 7 volts. For the 0.1 percent curve, care was taken to eliminate the effect of drift on the measurements.

Calculations predicting the errors arising from non-ideal behavior of transistors have not yet been made; however, the measurements seem to indicate that these are of a much smaller order than those resulting from typical transistor-pair mismatches.

Fig. 8 illustrates the multiplier used as a modulator.

B. Zero Suppression

Fig. 9 gives a plot of the output with one input grounded. The slope of the curves indicates a square-law relation, as predicted by (12). Symmetrical operation with respect to both inputs was noted. Deviations from

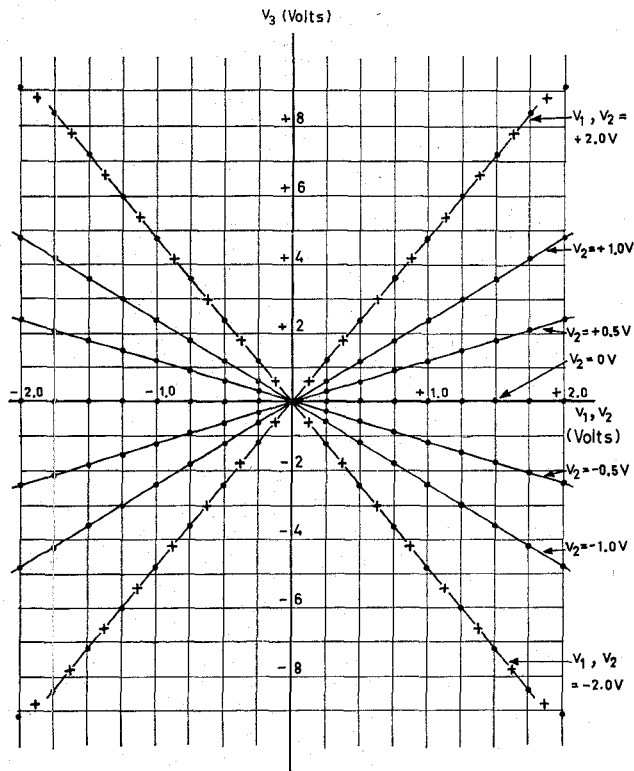


Fig. 6. Static linearity using pair 2. * v_1 varied, $v_2 =$ constant (0 volts, ± 0.5 volt, ± 1.0 volt, ± 2.0 volts); + v_2 varied, $v_1 =$ constant (± 2.0 volts).

this square-law and symmetrical behavior occurred for very well-matched transistor pairs (at extremely low error levels).

C. Frequency Response

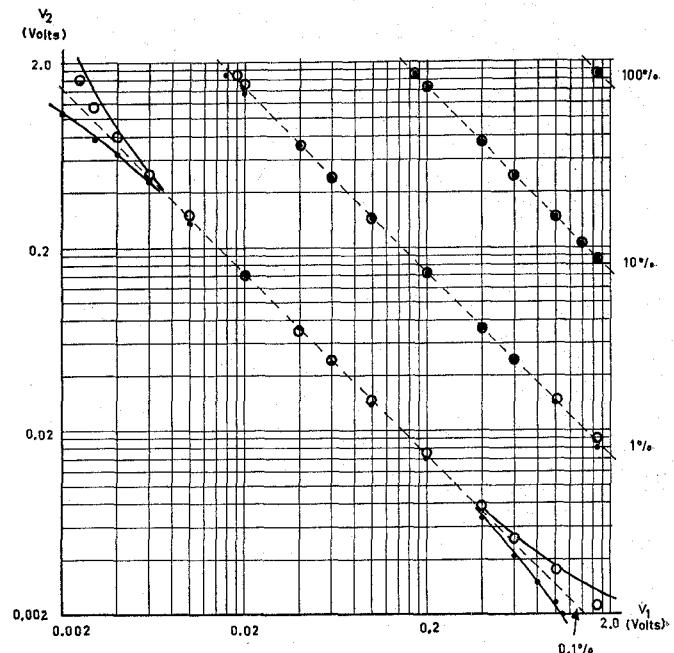
The bandwidth of such the multiplier stage itself is limited by the following three factors:

- 1) the parasitic capacitances to signal earth at the collectors of the current-fork transistors (these are the only nodes at which large voltage swings occurs);
- 2) the cut-off frequency (f_T) of the transistors;
- 3) the collector-base capacitances of the transistors.

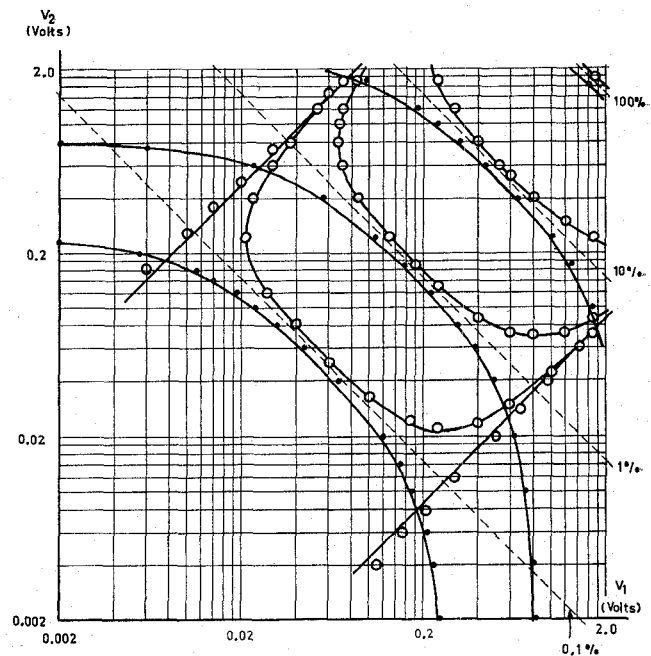
The 3-dB point of the multiplier (Fig. 5) was approximately 200 kHz (with respect to both inputs). Transistors having a higher f_T have been used in the design of a multiplier covering the video range (0–5 MHz); lower values of R_C and R_F and higher quiescent currents were used in this design; and a trade-off in the static linearity was noticed.

An exact analysis of transient response is complicated by the existence of base-collector capacitances of the current-fork transistors. At high-frequencies these capacitances and the extrinsic base resistances lead to an unbalance in the voltage drive across the intrinsic bases of the pairs of current-fork transistors. This unbalance gives rise to a nonlinear behavior of the multiplier and invalidates a frequency-domain analysis.

To reduce the effect of the base-collector capacitance on the linear behavior of the circuit at high frequencies it seems appropriate to include a unity-gain transistor



(a)



(b)

Fig. 7. Logarithmic plot of one input voltage as a function of the other input voltage when the output voltage is held constant at 7 volts (= 100 percent), 0.7 volt (= 10 percent), 0.07 volt (= 1 percent) and 0.007 volt (= 0.1 percent). (a) Pair 1; (b) pair 7. $\circ$ —measured values (first and third quadrants); $\bullet$ —measured values (second and fourth quadrants); — characteristics of ideal multiplier; — Curves calculated using (12) for (a) $(\Delta v - \Delta v') = 0.050$ mV and (b) $(\Delta v - \Delta v') = 4.49$ mV.

cascade stage at the collector of each of the current-fork transistors. This reduces the voltage swing at these collectors and minimizes the effect of the unwanted capacitances.

D. Drift

Drift at the output was measured with both inputs grounded. The sum of random short- and long-term and temperature (0–60°C) drifts was found to be of the order of 5 mV for a well-matched pair.

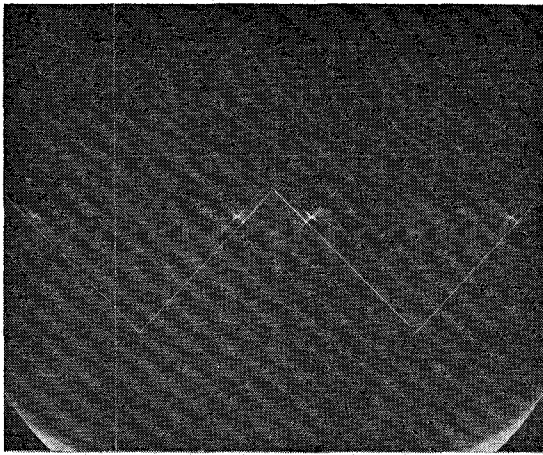


Fig. 8. Multiplier as a modulator. Vertical scale: 2 V/div. Horizontal scale: 1 ms/div. The frequencies are in the audio range.

E. Noise

The noise measured at the output with both inputs grounded was found to be less than 100 μ V rms over the bandwidth of interest.

F. Used as Squarer

The form of the unwanted terms in (12) suggests the suitability of the circuit to be used as a squarer. The expression describing the operation of the circuit as such takes the form:

$$i_{out} = -(2R_F/R_C I)(1 + \delta)v_{in}^2$$

where δ is directly proportional to the degree of mismatch and inversely proportional to absolute temperature; however, for a typical mismatch, $\delta \ll 1$.

Fig. 10 shows a dc plot of the multiplier when used as a squarer. Output voltages below 2 mV were difficult to measure due to the effects of superimposed random drift.

No deviation from symmetrical behavior was noticed when the polarity of the input voltage was reversed.

G. Alignment Procedure

It was found that a residual mismatch of transistors and a mismatch of resistors caused a dc offset at both inputs and at the output. The offsets at the inputs were eliminated by padding the emitter resistors of the balanced current drives. The output offset was remedied by appropriately adjusting the quiescent point of the output stage Q_5 , Q_6 .

Extremely close matching of the resistances R_C and R_F was found to be not necessary; in fact, the use of resistors in the 5 percent tolerance range was quite satisfactory. An expression indicating the small error arising from a finite mismatch of either R_C or R_F is given in Appendix III.

A particular project called for the construction of 40 multipliers and a batch of 90 dual transistors of type 2N2223A were obtained for this purpose. Measurement of the mismatch voltages of each pair indicated that 40

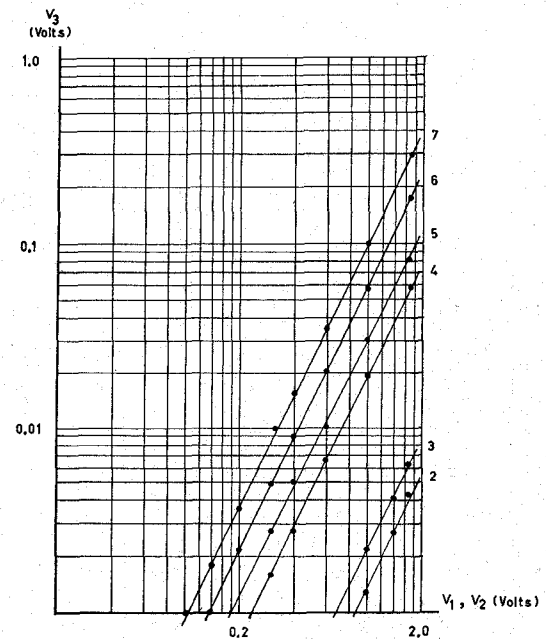


Fig. 9. Zero suppression. Behavior is in agreement with (12). Transistor pair number is indicated with each curve.

pairs of duals could be selected such that the values of $\Delta v - \Delta v'$ were less than 0.2 mV in all cases (a typical value was 0.1 mV). This degree of matching was sufficiently good, and eliminated the need for complex circuitry in each multiplier to compensate for the mismatch.

A group of more closely matched dual pairs could presumably be selected from a larger batch of dual pairs.

V. CONCLUSION

A transistor implementation of the feedback technique described has yielded a versatile four-quadrant analog multiplier¹ that overcomes many of the limitations of previous multipliers that make use of the unique characteristics of the bipolar transistor.

It is hoped that the design technique will fill a gap in the technology of nonlinear circuits by combining an accurate and predictable behavior with extreme simplicity of design and realization. Furthermore, the nature of the multiplier is such that it is easily cascaded to give an n -stage multiplier or a polynomial generator. The latter implementation is particularly attractive when powers of 2, 4, 8, etc., only are required, as the need for selecting matched pairs is eliminated (each stage performing a squaring function).

The realization of the multiplier in monolithic form, where a selection of matched pairs is technologically not feasible, would seem to limit its accuracy by the degree of matching obtained within the chip. Two possible techniques to overcome this limitation are

- 1) the application of an offset voltage between the transistor pairs to compensate for any mismatch

¹ Subject of Australian patent application 43033/69 and equivalent applications to the U. S., Japan, West Germany, and the United Kingdom.

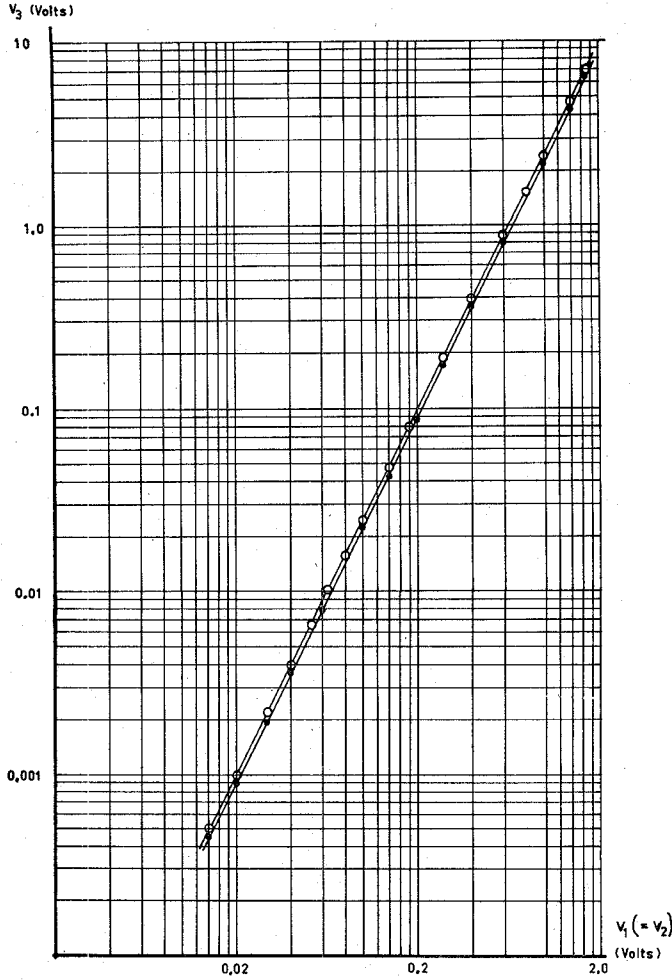


Fig. 10. Multiplier used as a squarer. Curves for positive and negative inputs are identical. ○—pair 2 ($\Delta v - \Delta v' = 0.10$ mV); ●—pair 7 ($\Delta v - \Delta v' = 4.49$ mV). Superimposed drift made measurements at output levels below 2 mV difficult.

voltage (a means of adjusting this voltage outside the chip would have to be provided);

- 2) the use of two multipliers as squarers in a quarter-square configuration; (this technique is valid where operation over a limited temperature range only is required).

Both of these techniques would involve additional circuit complexities.

APPENDIX I

Due to the symmetrical nature of the circuit, it can be simply shown that the signal components of the voltages at nodes A and B (Fig. 4) are equal in magnitude and opposite in phase. Assuming, then, that the signal voltage across the bases v_i , which is in fact analogous to χ_i in Section II, is negligibly small compared with other voltages appearing in the circuit, we may, by inspection, identify k_0 and k_F with circuit components in the following way:

$$k_0 = \frac{R_F}{R_C + R_F} \quad (15)$$

$$k_F = \frac{R_C}{2(R_C + R_F)}. \quad (16)$$

We have the following expressions for the output and the feedback currents:

$$I_i - I'_i = k_0(2\alpha_i - 1)(I_{i-1} - I'_{i-1}) + \epsilon_0 \quad (17)$$

$$i_F = k_F(2\alpha_i - 1) - \epsilon_F \quad (18)$$

where ϵ_0 and ϵ_F are errors due to the mismatched current forks and may be evaluated as follows.

$$\begin{aligned} \epsilon_0 &= k_0[(2\alpha_i - 1) - (2\alpha'_i - 1)] \frac{(I_{i-1} - I'_{i-1}) - I}{2} \\ &= k_0 \frac{(I_{i-1} - I'_{i-1}) - I}{2} \left[\tanh\left(\frac{-qv_i}{2KT}\right) \right. \\ &\quad + \left(\frac{q\Delta v}{2KT}\right) \operatorname{sech}^2\left(\frac{-qv_i}{2KT}\right) \\ &\quad + \left(\frac{q\Delta v}{2KT}\right)^2 \operatorname{sech}\left(\frac{-qv_i}{2KT}\right) \tanh\left(\frac{-qv_i}{2KT}\right) + \dots \\ &\quad \left. - \tanh\left(\frac{-qv_i}{2KT}\right) - \left(\frac{q\Delta v'}{2KT}\right) \operatorname{sech}^2\left(\frac{-qv_i}{2KT}\right) - \dots \right] \\ &\approx k_0 \frac{q(\Delta v - \Delta v')}{4KT} [(I_{i-1} - I'_{i-1}) - I] \left[1 - \left(\frac{qv_i}{2KT}\right)^2 \right] \end{aligned} \quad (19)$$

for $\Delta v, \Delta v' \ll 2KT/q$, $v_i < 2KT/q$ and neglecting higher order terms. Similarly,

$$\epsilon_F \approx k_F \frac{q(\Delta v - \Delta v')}{4KT} [(I_{i-1} - I'_{i-1}) - I] \left[1 - \left(\frac{qv_i}{2KT}\right)^2 \right]. \quad (20)$$

Also, for $v_i < 2KT/q$ we have

$$2\alpha_i - 1 \approx \frac{q(v_i + \Delta v)}{2KT} \quad (21)$$

and

$$v_i + \Delta v \approx (i_i - i_F) \frac{4\beta KT}{qI} \quad (22)$$

The term $4\beta KT/qI$ in (22) represents the balanced small-signal resistance seen across the pairs of bases when the currents through all of the current-fork transistors are approximately equal. It shall be assumed for the sake of simplicity that the β values of the current-splitting transistors are equal and large.

By combining (17), (18), (21), and (22) we obtain an expression for the output:

$$\begin{aligned} I_i - I'_i &= \frac{\gamma k_0}{k_F I} (I_{i-1} - I'_{i-1}) i_i + \frac{\gamma k_0}{k_F I} (I_{i-1} - I'_{i-1}) \epsilon_F + \epsilon_0 \\ &= \frac{\gamma k_0}{k_F I} (I_{i-1} - I'_{i-1}) i_i + \epsilon_T \end{aligned} \quad (23)$$

where

$$\gamma = \frac{1}{1 + \frac{1}{2\beta k_F}} \approx 1$$

for typical values of k_F and β .

In order to derive expressions (in terms of input variables) for the error terms ϵ_0 and ϵ_F as given by (19) and (20) we may assume that v_i is given by:

$$v_i = i_i r_{in} \quad (24)$$

where r_{in} is the small-signal input impedance to the multiplier as seen by the current source i_i . It may be shown that:

$$r_{in} \approx \frac{2KT}{qIk_F}$$

where we obtain:

$$v_i \approx \frac{2KT i_i}{qIk_F} \quad (25)$$

By substituting (19), (20), and (25) in (23) and expressing the input variables in their normalized form we obtain the desired expression for the normalized output:

$$Z = \gamma' \left[XY + \frac{q(\Delta v - \Delta v')}{4KT} (X^2 + Y^2 - X^2 Y^2 - 1) \right] \quad (26)$$

where

$$\gamma' = \gamma \left(\frac{1}{1 + \frac{1}{\beta}} \right)$$

(i.e., γ' takes account of the currents lost at the collectors of the transistors due to a finite value of β). Under normal operating conditions the magnitude of the term $X^2 Y^2$ is much smaller than the other terms and may be neglected.

APPENDIX II

The work of Gilbert [7] has led to a four-quadrant transistor analog multiplier that bears some resemblance to the one presented in this paper. In order to obtain a quantitative comparison of the performance of the two types, the analogous expression to (26) may be obtained for the Gilbert multiplier by combining (6), (14), and (15) ([7], pp. 366-367), to give:

$$Z = XY + \frac{q\Delta \Delta v_1}{4KT} (X^2 Y - Y + X^2 - 1) + \frac{q\Delta \Delta v_2}{4KT} (X^2 Y - Y - X^2 + 1) \quad (27)$$

where

$$\frac{q\Delta \Delta v_1}{4KT} \approx (1 - \gamma_1)$$

and

$$\frac{q\Delta \Delta v_2}{4KT} \approx (1 - \gamma_2).$$

A comparison of (26) and (27) reveals explicitly the difference in the nature of the error terms (due to area mismatch) between the Gilbert multiplier and the one presented in this paper. Furthermore, the derived error terms will determine which of the multipliers is most suitable for use in a particular systems configuration.

APPENDIX III

In order to examine the effect of a mismatch of the resistances R_C and R_F the output error due to small increments Δk_0 and Δk_F of the values of k_0 , and k_F at each of the collectors of the four current-fork transistors shall be calculated. For the circuit configuration of Fig. 4 we have

$$k_0 = \frac{R_F}{R_C + R_F}$$

and

$$k_F = \frac{R_C}{2(R_C + R_F)};$$

therefore, $2k_F + k_0 = 1$ and $\Delta k_0 = -2\Delta k_F$. Let Δk_1 , Δk_2 , Δk_3 , and Δk_4 be the increments Δk_0 ($= -2\Delta k_F$) for the four collector circuits due to finite errors in either R_C or R_F .

Equation (23) may now be used to compute the total normalized error $\epsilon_{TN} - (\epsilon_T/k_0 I)$ in terms of the resistance mismatch parameters. Therefore, assuming a unity value for γ' , we have

$$\begin{aligned} \epsilon_{TN} = & \frac{1}{4k_0} \left[1 - \frac{k_0}{2k_F} X^2 \right] [Y(\Delta k_1 + \Delta k_2 - \Delta k_3 - \Delta k_4) \\ & + \Delta k_1 - \Delta k_2 + \Delta k_3 - \Delta k_4] \\ & + \frac{X}{4k_0} \left[1 + \frac{k_0}{2k_F} \right] [Y(\Delta k_1 + \Delta k_2 - \Delta k_3 - \Delta k_4) \\ & + \Delta k_1 - \Delta k_2 - \Delta k_3 + \Delta k_4]. \end{aligned} \quad (28)$$

In this expression, the error terms having the variable factors X and Y can be easily compensated in a practical realization. Moreover, the term containing the variable factor XY merely modifies the constant of multiplication. The term that gives rise to an essential nonideal behavior of the multiplier is

$$\begin{aligned} \epsilon'_{TN} = & \frac{X^2}{8k_F} [Y(\Delta k_1 + \Delta k_2 - \Delta k_3 - \Delta k_4) \\ & + \Delta k_1 - \Delta k_2 + \Delta k_3 - \Delta k_4]. \end{aligned} \quad (29)$$

The condition that must be obeyed to reduce this term to zero is $\Delta k_1 = \Delta k_4$ and $\Delta k_2 = \Delta k_3$. From this we can infer that such a multiplier will not suffer nonlinear distortion with respect to either input if the resistance values R_C and R_F are matched for the first and fourth collectors and for the second and third collectors.

To illustrate the magnitude of the coefficient of the error term arising from a typical resistance mismatch let

us assume an error of 1 percent in the value of R_F at the collector of only the first transistor of a multiplier for which $k_0 = k_F = 0.5$. In this case we have $\Delta k_1/k_F = \frac{1}{4}$ percent and $\Delta k_2 = \Delta k_3 = \Delta k_4 = 0$; therefore,

$$\epsilon'_{NT} = \frac{X^2(Y+1)}{3200}.$$

The same result is obtained for a 1 percent change in the value of R_G .

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Method of Improving Switching Speed

Abstract—The saturation characteristic of a switching transistor can be improved by using, for example, gold doping, buried layers, or clamp circuits. However, some important factors such as switching speed, loading capability, cost, and reliability may have to be sacrificed. This correspondence describes a new technique to significantly improve not only the saturation characteristic, but also the switching speed by utilizing a two-collector-terminal transistor (TCT). The TCT structure and theory are presented. Results of experiments using the TCT and the conventional transistor are compared.

INTRODUCTION

The use of gold doping [1] to kill the lifetime of the carriers is one of the techniques used to improve device saturation characteristics. The disadvantages of this technique are low transistor beta, high leakage currents, and low switching speed. Therefore it is not suitable to use a gold-doped transistor in a high-speed logic circuit. The use of additional components to limit the base overdrive of a switching transistor [2] and to control the saturation level of transistor-transistor logic (TTL) and diode-transistor logic (DTL) gates [3] is another technique to improve switching speed. Additional components required in the input [2] and output [3] may thus decrease the loading capability. Furthermore, cost and reliability may also be affected. Reducing R_{CS} (collector series resistance) by increasing the sizes of the heavily doped buried layer and the collector contact is the conventional way of improving saturation. However, as the area increases, the stray capacitance increases. Diode clamping circuits [4], [5] connected to the collector are also a well-known method of improving saturation, but the transistor's switching speed in the active region may be sacrificed. This correspondence will describe a new technique

to improve the saturation characteristic and switching speed by utilizing a two-collector-terminal transistor (TCT) without having the drawbacks mentioned previously.

STRUCTURE AND THEORY

Fig. 1 shows the structure of a TCT transistor. The primary collector C_1 and secondary collector C_2 were separated by scratching away part of the metallization of a conventional structure and rebounding, shown in Fig. 2, for use as an experimental TCT transistor. In the design of an integrated circuit, C_1 and C_2 can be arranged such that they will be topologically convenient for interconnection. Conventional photolithographic techniques and diffusion processes can be used to fabricate this device. C_1 is connected to a modified collector load resistance R_C , and C_2 to the base of the next stage (with a high input impedance) such as the emitter follower of a CSEF logic. As a result of this configuration, the collector current I_C will flow through C_1 , and the input current to the next stage I_{B2} will flow through C_2 , as shown in Fig. 3(a). The collector series resistance R_{CS1} now can be used as part of the load resistance R_C , so that the voltage drop $I_C R_{CS1}$ becomes part of the output ($I_C R_C + I_C R_{CS1}$), instead of being a drop of the input saturation voltage $V_{IN(sat)}$, which is defined as the input up-level voltage at 0.6-ns additional saturation delay, where 0.6 ns is arbitrarily chosen from a system viewpoint.

For a conventional structure shown in Fig. 3(b), $V_{IN(sat)}$ is

$$V_{IN(sat)conv} = V_{CC} - I_C R_C - I_{C'} R_{CS} + V_{BC'} \quad (1)$$

where

$$I_C = I_{C'} + I_{B2}$$

$$I_{C'} = \frac{h_{FE}}{1 + h_{FE}} \cdot I_B.$$

For the TCT structure shown in Fig. 3(a), the $V_{IN(sat)}$ becomes